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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/093,950	01/06/2023	Yu Fang	SONA-0012-U01-C01	3517

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EXAMINER
JHA, ABDHESH K

ART UNIT	PAPER NUMBER
3668	

NOTIFICATION DATE	DELIVERY MODE
09/07/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/093,950	Applicant(s) Fang et al.	
	Examiner ABDHESH K JHA	Art Unit 3668	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 08/22/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-13 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-8 and 10-13 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 01/06/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 08/22/2023.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. Claims 1-13 are considered in this office action. Claim 9 is cancelled. Claims 1-8, 10-13 are pending examination.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 103

3. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

6. Claims 1-8, and 10-13 are rejected under 35 U.S.C. 103 as being unpatentable over Sugaiwa et al. (US2018/0267530) and in view of Han et al. (US2018/0076970) and herein after will be referred as Sugaiwa and Han respectively.

Regarding Claim 1, Sugaiwa teaches **a system** (Para [0032] Fig .1 Line 1-3 #100 an automated driving system: "FIG. 1 is a block diagram showing the configuration of an automated driving system **100** of a first embodiment."), **comprising:**

a vehicle ("Para [0032]: "As shown in FIG. 1, the automated driving system **100** is installed on a vehicle V, such as a passenger automobile.") **having a first network comprising a first plurality of end points** (Para [0033] : "The automated driving system **100** includes an external sensor or sensors **3**, GPS (Global Positioning System) receiver **4**, internal sensor or sensors **5**, map database **6**, navigation system **7**, actuators **8**, input device **9**, and an ECU (Electronic Control Unit) **10**." Here actuators are being interpreted as end points of the vehicle V.);

a controller (Para [0041]: "The ECU **10** controls the vehicle V. The ECU **10** is an electronic control unit having CPU (Central Processing Unit), ROM (Read Only Memory), RAM (Random Access Memory), CAN (Controller Area Network) communication circuit, and so forth.,") **comprising: an automation definition circuit structured to interpret an automation description** (Para [0053-0054]: "If a negative decision (NO) is obtained in step **S2**, the control

proceeds to step S6 that will be described later. If an affirmative decision (YES) is obtained in step S2, the obstacle recognizing unit 11 recognizes the obstacle based on the detection result of the external sensor(s) 3 (step S3). Then, it is determined whether overwriting input is being entered from the user into the input device 9 (step S4). If an affirmative decision (YES) is obtained in step S4, the overwriting unit 14 executes overwriting operation to exclude the obstacle from the obstacle recognition result obtained by the obstacle recognizing unit 11 (step S5)." Here overwriting input / users' input is being interpreted as automation description).

an automation management circuit structured to provide an automated action plan in response to the automation description (Para [0054-0055]: "Namely, in step S5, the obstacle recognition result is overwritten into a result indicating that no obstacle was recognized. After execution of step S5, the overwriting unit 14 stores a log indicating execution of the overwriting operation and the content of the overwriting operation, and generates the obstacle recognition result subjected to the overwriting operation, to the traveling plan creating unit 12. Then, the control proceeds to step S6 that will be described later. On the other hand, if a negative decision (NO) is obtained in step S4, the overwriting unit 14 does not execute overwriting operation, and the obstacle recognition result obtained by the obstacle recognizing unit 11 is generated as it is to the traveling plan creating unit 12."……**[0055]** "In step S6, the traveling plan creating unit 12 creates a traveling plan, based on at least one of the detection results of the external sensors 3 and the internal sensors 5, position information of the GPS receiver 4, map information of the map database 6, various kinds of information of the navigation system 7, and the obstacle recognition result.");

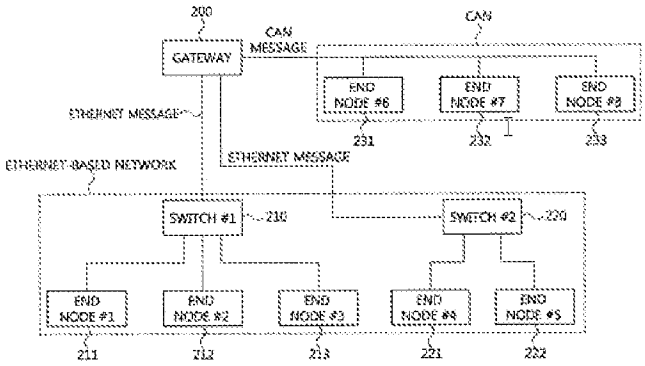
and an automation execution circuit structured to provide an automation command in response to the automated action plan (Para [0055]: "Then, the vehicle controller **13** performs vehicle control based on the traveling plan thus created, and causes the vehicle V to autonomously travel according to the traveling plan.")**;**

and wherein a first end point, of the first plurality of end points is responsive to the automation command to implement an automated vehicle response (Para [0047]: "The vehicle controller **13** transmits control signals to the actuators **8**, based on the traveling plan created by the traveling plan creating unit **12**. In this manner, the vehicle V is controlled, so that the vehicle V autonomously travels according to the traveling plan.").

Han teaches a second network comprising a second plurality of end points; and a second end point, of the second plurality of end points (Fig 2 as produced below: here second network is ethernet based network and second end point is end node #2).

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FIG. 2



It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified Sugaiwa to incorporate the teachings of Han to include a second network and a second end points. Doing so would optimize the vehicle operations.

Regarding Claim 2, Sugaiwa in view of Han teaches the system of claim 1.

Sugaiwa also teaches **wherein the automation description comprises at least a portion of a policy for the vehicle** (Para [0054] Line 7-11: "After execution of step S5, the overwriting unit 14 stores a log indicating execution of the overwriting operation and the content of the overwriting operation, and generates the obstacle recognition result subjected to the overwriting operation, to the traveling plan creating unit 12.").

Regarding Claim 3, Sugaiwa in view of Han teaches the system of claim 1.

Sugaiwa also teaches **wherein the automated action plan comprises a data description of data to be collected** (Para [0052]: "Initially, the detection result of the external sensor(s) 3, namely, information concerning ambient surroundings of the vehicle V, is obtained (step S1).").

Regarding Claim 4, Sugaiwa in view of Han teaches the system of claim 1.

Sugaiwa also teaches **wherein the end point of the plurality of end points comprises an actuator of the vehicle** (Para [0033]: "The automated driving system 100 includes an external sensor or sensors 3, GPS (Global Positioning System) receiver 4, internal sensor or sensors 5, map database 6, navigation system 7, actuators 8, input device 9, and an ECU (Electronic Control Unit) 10.").

Regarding Claim 5, Sugaiwa in view of Han teaches the system of claim 1.

Sugaiwa also teaches wherein **the first end point of the first plurality of end points comprises a sensor of the vehicle** (Fig.1 #3 Para [0033]: "The automated driving system 100 includes an external sensor or sensors 3, GPS (Global Positioning System) receiver 4, internal sensor or sensors 5, map database 6, navigation system 7, actuators 8, input device 9, and an ECU (Electronic Control Unit) 10.").

Regarding Claim 6, Sugaiwa in view of Han teaches the system of claim 1.

Sugaiwa also teaches wherein **the first end point of the first plurality of end points comprises a second controller** (Para [0041]: "The ECU 10 may consist of two or more electronic control

units. The ECU 10 includes an obstacle recognizing unit 11, a traveling plan creating unit 12, a vehicle controller 13, and an overwriting unit 14.)

Regarding Claim 7, Sugaiwa in view of Han teaches the system of claim 1.

Sugaiwa also teaches wherein **the second controller is further configured to provide a confirmation value to the controller in response to implementing the automated vehicle response** (Para [0054]: "Namely, in step S5, the obstacle recognition result is overwritten into a result indicating that no obstacle was recognized. After execution of step S5, the overwriting unit 14 stores a log indicating execution of the overwriting operation and the content of the overwriting operation, and generates the obstacle recognition result subjected to the overwriting operation, to the traveling plan creating unit 12." Here vehicle autonomously traveling and not colliding according to traveling plan is interpreted as confirmation value to the control in response to the automated vehicle response.).

Regarding Claim 8, Sugaiwa in view of Han teaches the system of claim 1.

Sugaiwa also teaches further comprising **a third end point of the first plurality of end points, wherein the third end point is responsive to the automation command to implement the automated vehicle response** (Para [0039]: "The actuators 8 are devices that perform traveling control of the vehicle V. The actuators 8 include at least an engine actuator, a brake actuator, and a steering actuator. The engine actuator changes the amount of air supplied to the engine (e.g., changes the throttle opening) according to a control signal from the ECU 10, thereby to control the driving force of the vehicle V. In the case where the vehicle V is a hybrid vehicle or an electric vehicle, the engine actuator controls driving force of a motor as a power source. The

brake actuator controls a brake system according to a control signal from the ECU **10**, so as to control braking force applied to the wheels of the vehicle V. As the brake system, a hydraulic brake system may be used, for example. In the case where the vehicle V is equipped with a regenerative brake system, the brake actuator may control both the hydraulic brake system and the regenerative brake system. The steering actuator controls driving of an assist motor that controls steering torque in an electric power steering system, according to a control signal from the ECU **10**. In this manner, the steering actuator controls steering torque of the vehicle V." here engine actuator, brake actuator and steering actuators are being interpreted as one of the second end points).

Regarding Claim 10, Sugaiwa in view of Han teaches the system of claim 1.

Sugaiwa also teaches **wherein the automated action plan comprises an event detection description, and wherein the automation execution circuit is further structured to detect an event in response to the event detection description, and to implement the automated vehicle response further in response to the detected event** (Para [0053-0055]: "If a negative decision (NO) is obtained in step S**2**, the control proceeds to step S**6** that will be described later. If an affirmative decision (YES) is obtained in step S**2**, the obstacle recognizing unit **11** recognizes the obstacle based on the detection result of the external sensor(s) **3** (step S**3**). Then, it is determined whether overwriting input is being entered from the user into the input device **9** (step S**4**) .....If an affirmative decision (YES) is obtained in step S**4**, the overwriting unit **14** executes overwriting operation to exclude the obstacle from the obstacle recognition result obtained by the obstacle recognizing unit **11** (step S**5**). Namely, in step S**5**, the obstacle recognition result is overwritten into a result indicating that no obstacle was recognized.

After execution of step S5, the overwriting unit 14 stores a log indicating execution of the overwriting operation and the content of the overwriting operation, and generates the obstacle recognition result subjected to the overwriting operation, to the traveling plan creating unit 12. Then, the control proceeds to step S6 that will be described later. On the other hand, if a negative decision (NO) is obtained in step S4, the overwriting unit 14 does not execute overwriting operation, and the obstacle recognition result obtained by the obstacle recognizing unit 11 is generated as it is to the traveling plan creating unit 12..... In step S6, the traveling plan creating unit 12 creates a traveling plan, based on at least one of the detection results of the external sensors 3 and the internal sensors 5, position information of the GPS receiver 4, map information of the map database 6, various kinds of information of the navigation system 7, and the obstacle recognition result. Then, the vehicle controller 13 performs vehicle control based on the traveling plan thus created, and causes the vehicle V to autonomously travel according to the traveling plan.")

Regarding Claim 11, Sugaiwa in view of Han teaches the system of claim 1. Sugaiwa also teaches wherein **the automation description further comprises a vehicle operating condition, and wherein the automation management circuit is further structured to provide the automated action plan in response to the vehicle operating condition** (Para [0036]:"
The internal sensors 5 are detection devices that detect traveling conditions of the vehicle V (i.e., how the vehicle V is operated). The internal sensors 5 include at least a vehicle speed sensor. The vehicle speed sensor is a detection device that detects the speed of the vehicle V. As the vehicle speed sensor, a wheel speed sensor that is provided on a wheel of the vehicle V or a driveshaft, or the like, that rotates as a unit with the wheel, for detecting the rotational speed of

the wheel, is used. The vehicle speed sensor transmits the detected vehicle speed information to the ECU **10**." And Para [0055]: "In step S**6**, the traveling plan creating unit **12** creates a traveling plan, based on at least one of the detection results of the external sensors **3** and the internal sensors **5**, position information of the GPS receiver **4**, map information of the map database **6**, various kinds of information of the navigation system **7**, and the obstacle recognition result. Then, the vehicle controller **13** performs vehicle control based on the traveling plan thus created, and causes the vehicle V to autonomously travel according to the traveling plan.").

Regarding Claim 12, Sugaiwa teaches the system of claim 1. Sugaiwa also teaches **wherein the automation description further comprises an external operating condition, and wherein the automation management circuit is further structured to provide the automated action plan in response to the external operating condition** (Para [0034]: "The external sensor **3** is a detection device that detect ambient surroundings (external conditions) as circumstances surrounding the vehicle V. The external sensor **3** includes at least one of a camera and a radar sensor. The camera is an image pickup device that captures an image of the ambient surroundings. The camera is provided on the rear side of the front windshield of the vehicle V. The camera transmits captured image information to the ECU **10**. "And Para [0055]: "In step S**6**, the traveling plan creating unit **12** creates a traveling plan, based on at least one of the detection results of the external sensors **3** and the internal sensors **5**, position information of the GPS receiver **4**, map information of the map database **6**, various kinds of information of the navigation system **7**, and the obstacle recognition result. Then, the vehicle controller **13** performs vehicle control based on the traveling plan thus created, and causes the vehicle V to autonomously travel according to the traveling plan.").

Regarding Claim 13, Sugaiwa in view of Han teaches the system of claim 1.

Sugaiwa also teaches **wherein the automated action plan comprises at least one of an actuator response or a trigger description** (Para [0047])

Response to Arguments

7. Applicant's arguments with respect to claims 1-8,10- 13 have been considered but are moot because the new ground of rejection does not rely on any reference applied in the prior rejection of record for any teaching or matter specifically challenged in the argument. The IDS filed on 08/22/2023 was considered and Han et al which was listed on the IDS is being used to teach the amended claim limitations of independent claim 1.

8. Independent Claim 1 was originally rejected under 102 however in view of Han et al. which was listed in newly filed IDS and in view of amendments to the claim 1, now currently amended claim 1 is being rejected under 103 Sugaiwa in view of Han as explained in detail above. Han teaches the second network and second end points as clearly disclosed in Fig.2.

9. Additionally, applicant simply state that newly amended claim 3 is not being taught by Sugaiwa. However, in view of new amendments, the other portion of Sugaiwa reference teaches the amended claim limitation as explained in detail above.

Next the applicant argues that claim 7 is not being taught by the reference Sugaiwa. Just to provide the high customer service the applicant is citing different portion of the prior art Sugaiwa which clearly teaches about providing confirmation value Para [0054] After execution of step S5, the overwriting unit 14 stores a log indicating execution of the overwriting operation and the

content of the overwriting operation, and generates the obstacle recognition result subjected to the overwriting operation, to the traveling plan creating unit 12." Here vehicle autonomously traveling and not colliding according to traveling plan is interpreted as confirmation value to the control in response to the automated vehicle response.

10. The examiner believes he has responded to all the arguments presented by the applicant at this time. However, if the applicant believes that the examiner has missed any arguments to respond, the applicant is invited to call the examiner directly to expedite the process.

Conclusion

11. Applicant's submission of an information disclosure statement under 37 CFR 1.97(c) with the fee set forth in 37 CFR 1.17(p) on 08/22/2023 prompted the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 609.04(b). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ABDHESH K JHA whose telephone number is (571)272-6218. The examiner can normally be reached M-F:0800-1700.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, James J Lee can be reached on 571-270-5965. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/ABDHESH K JHA/
Primary Examiner, Art Unit 3668